Customer Churn Prediction

Business insights report | 2026-10-11 | model: Logistic Regression

1,500 Customers	**755** Flagged high-risk	**$53,612** Revenue at risk / month
$643,347 Revenue at risk / year	**0.825** ROC-AUC (test)	**0.825** Recall (test)

Key findings

- Overall churn rate is 26.5% (398 of 1,500 customers); the classes are imbalanced, so accuracy alone is misleading.
- Contract type: 'Month-to-month' customers churn at 42.6% versus 2.5% for 'Two year'.
- Internet service: 'Fiber optic' customers churn at 41.5% versus 7.9% for 'No'.
- Payment method: 'Electronic check' customers churn at 44.4% versus 16.4% for 'Credit card (automatic)'.
- Tenure: '0-6 months' customers churn at 56.0% versus 7.4% for '49+ months'.
- Tech support: 'No' customers churn at 40.9% versus 15.7% for 'Yes'.
- Online security: 'No' customers churn at 41.6% versus 14.6% for 'Yes'.

- Precision 0.462, F1 0.592 at threshold 0.36.
- Estimated campaign net benefit: $62,025 per year (save rate 30%, offer cost $40).

Statistical associations are not proof of causation.

Model comparison (cross-validation, training data)

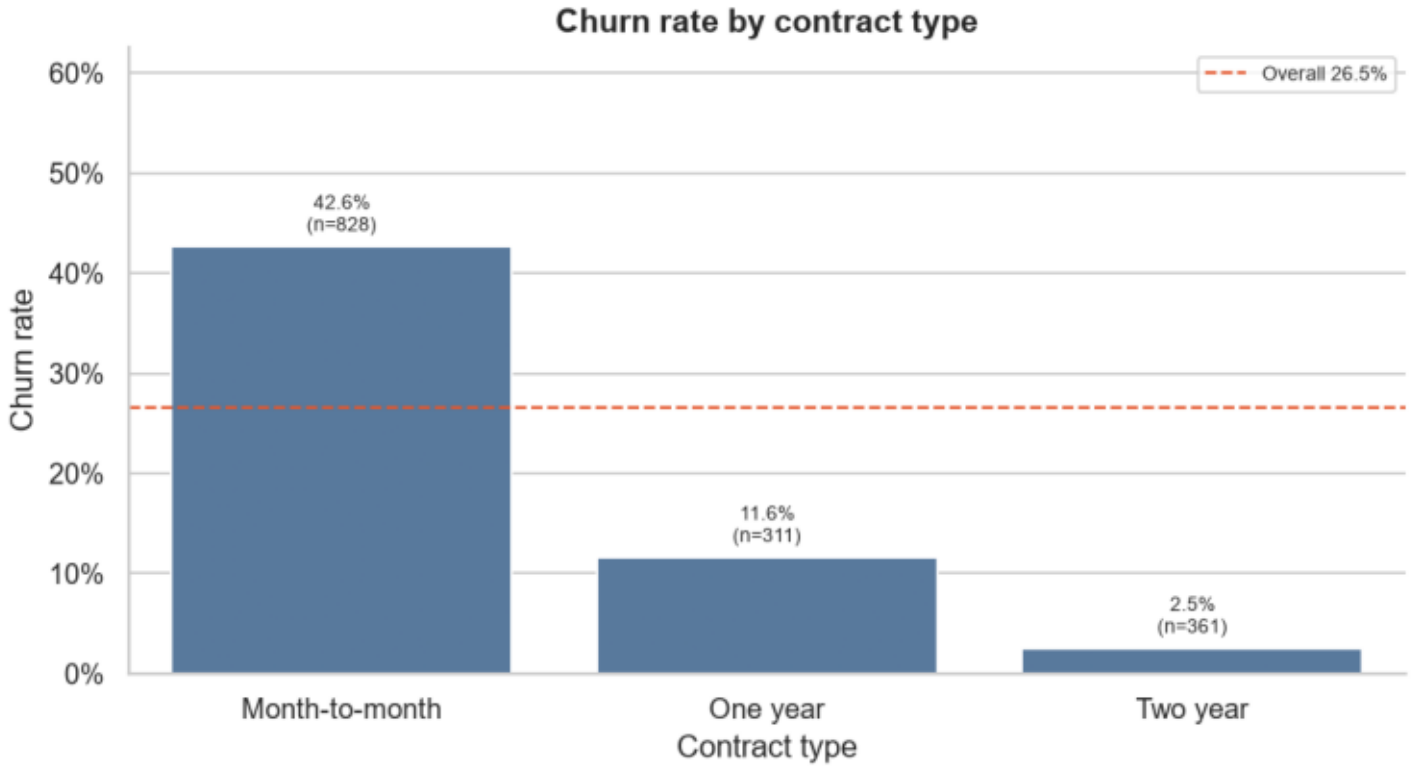
display_name	cv_roc_auc	cv_pr_auc	cv_recall	cv_precision	cv_f1	selected
Logistic Regression	0.847	0.657	0.789	0.520	0.626	True
Random Forest	0.847	0.663	0.752	0.522	0.616	False
Gradient Boosting	0.843	0.653	0.522	0.651	0.578	False
XGBoost	0.848	0.660	0.799	0.528	0.636	False

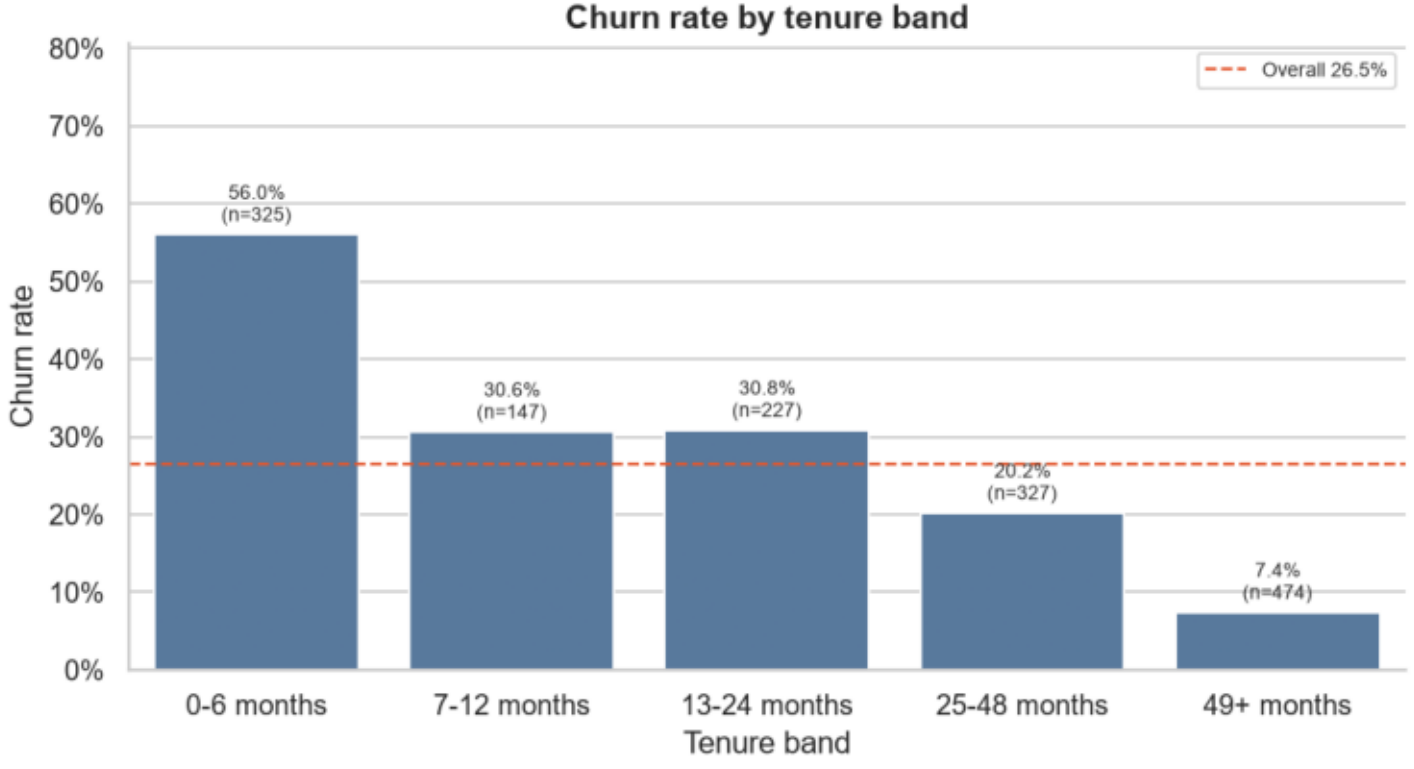
Segments needing retention action

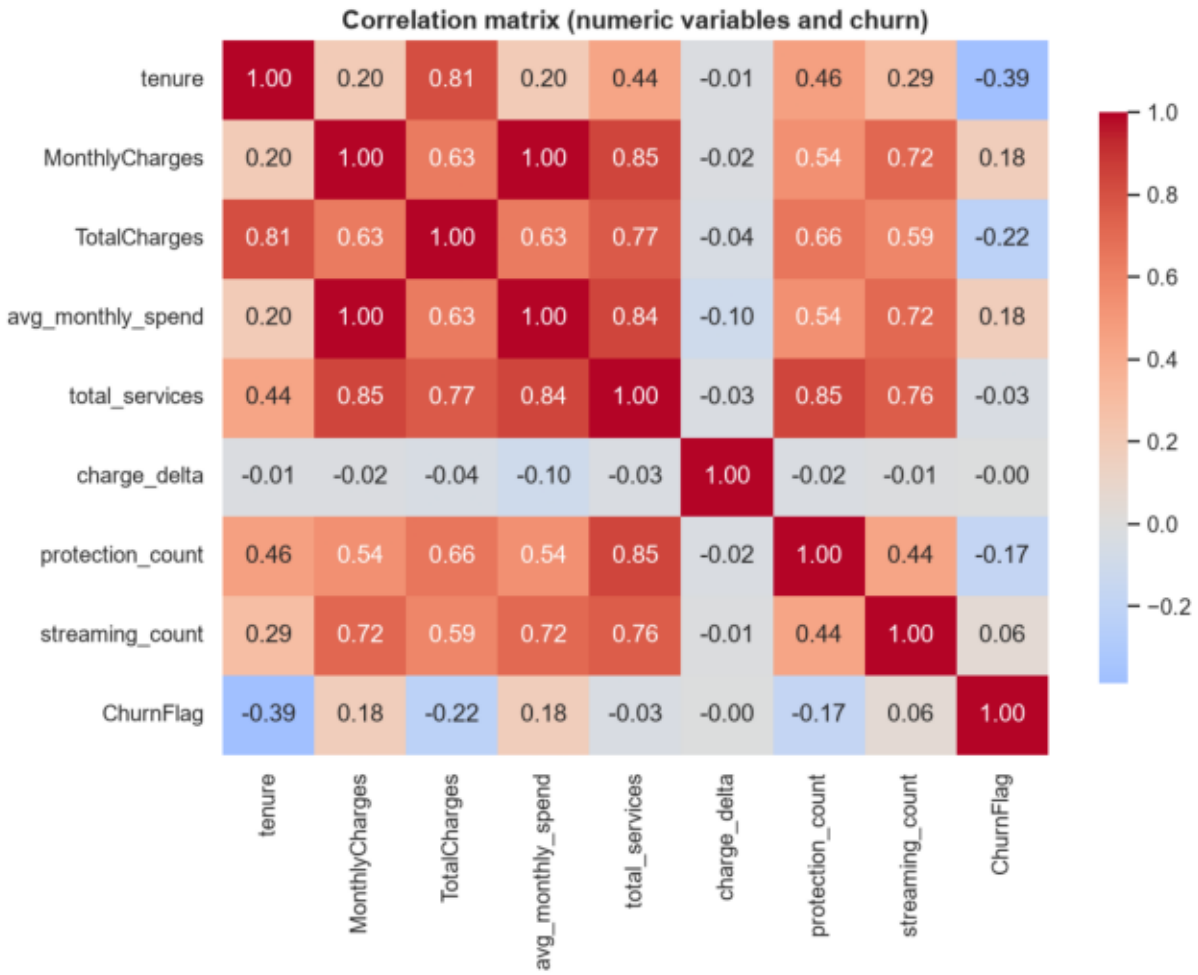
segment	customers	flagged_pct	monthly_revenue_at_risk	urgency
Month-to-month | 0-6 month	305	0.987	17,236.900	Urgent
Month-to-month | 13-24 mo	175	0.863	11,510.350	Urgent
Month-to-month | 25-48 mo	163	0.718	9,974.150	Urgent
Month-to-month | 7-12 mont	117	0.803	6,627.000	Urgent
Month-to-month | 49+ montl	68	0.588	3,624.450	Urgent
One year | 49+ months	134	0.127	1,661.900	Monitor
One year | 25-48 months	108	0.167	1,652.950	Monitor
One year | 13-24 months	33	0.212	621.000	Monitor
One year | 0-6 months	16	0.312	299.450	High
One year | 7-12 months	20	0.200	295.200	Monitor

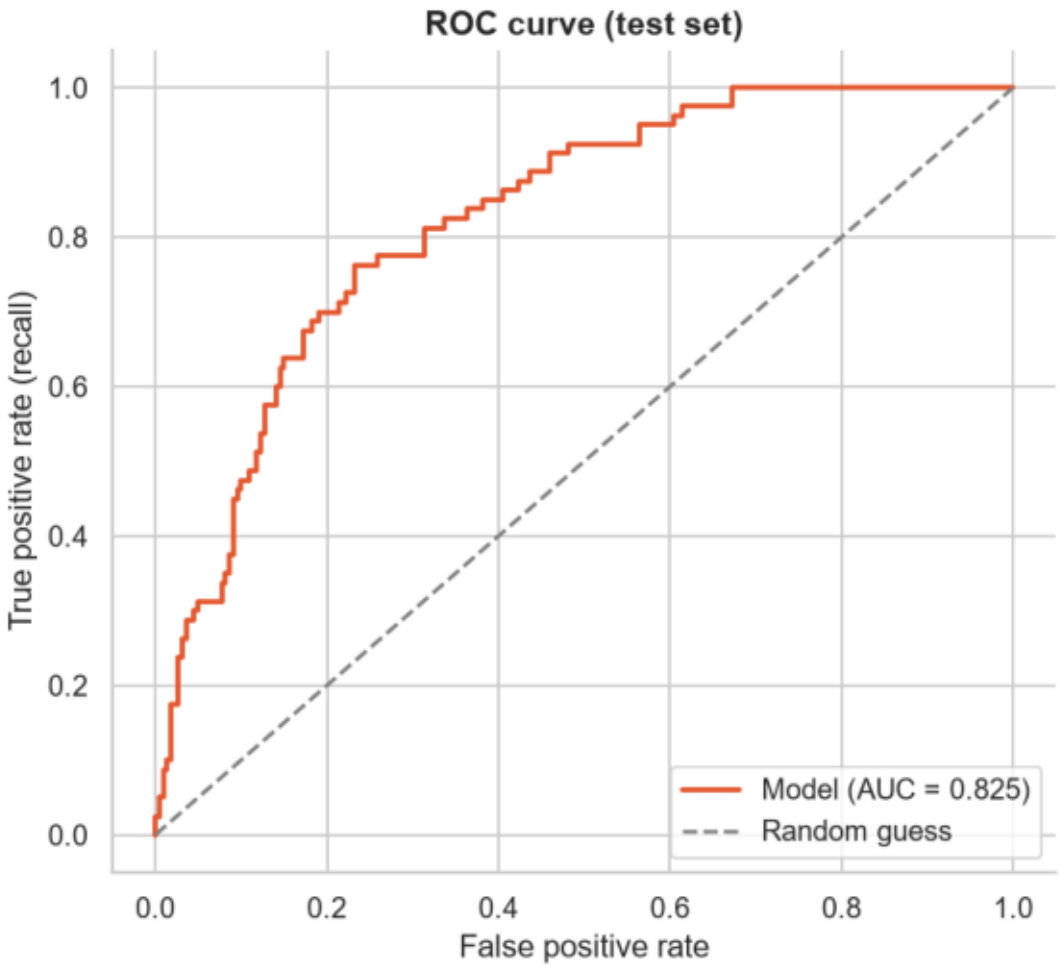
Prioritised retention actions

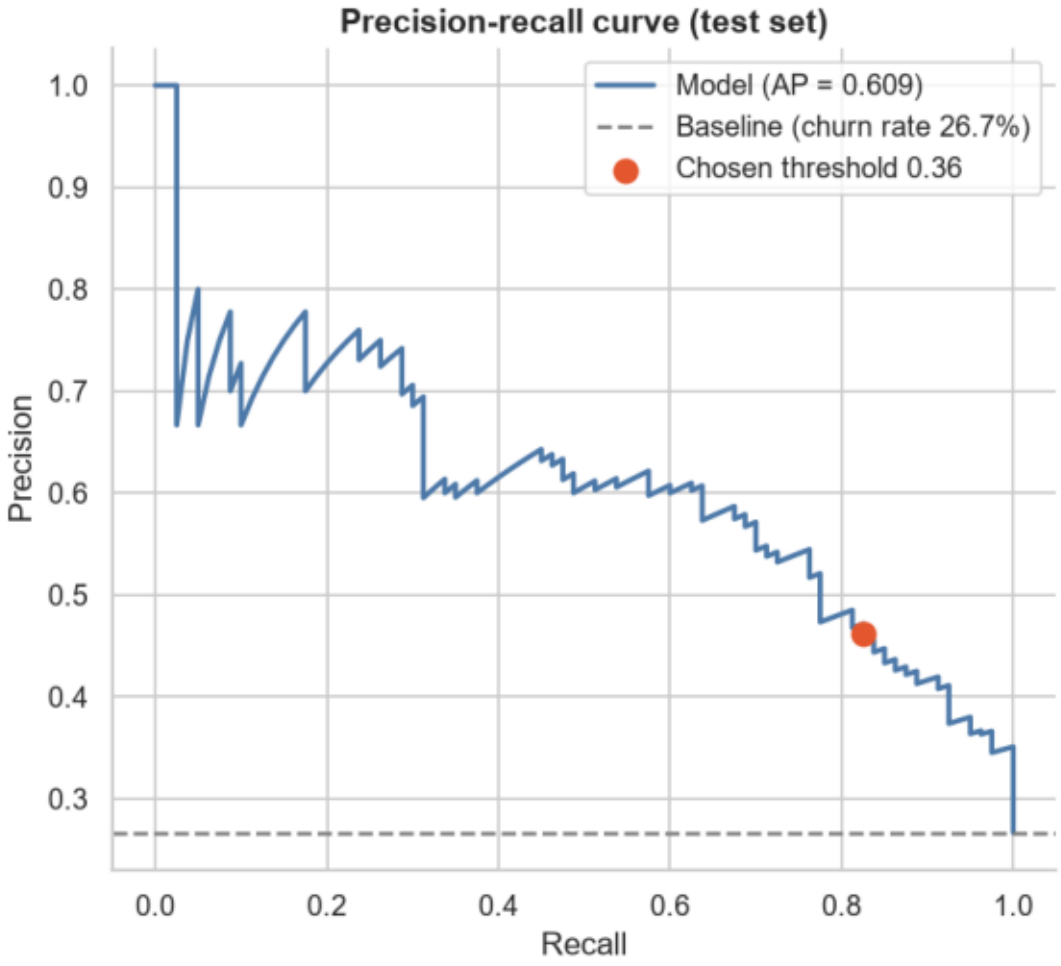
recommended_action	impact	flagged_customers_affected	monthly_revenue_covered
Offer a loyalty upgrade to a 1- [...]	3	308	25,108.950
Offer a 12-month contract with a [...]	3	395	23,863.900
Schedule a 30-day onboarding [...]	3	306	17,536.350
Offer a free 3-month Tech [...]	2	476	35,433.250
Offer a one-off bill credit for [...]	2	405	30,790.450
Run a service-quality check and [...]	2	319	29,891.350
Proactive bill review: explain [...]	2	51	3,969.300
Offer assisted support: priority [...]	1	183	14,531.500
Bundle security/backup add-ons [...]	1	148	9,119.900

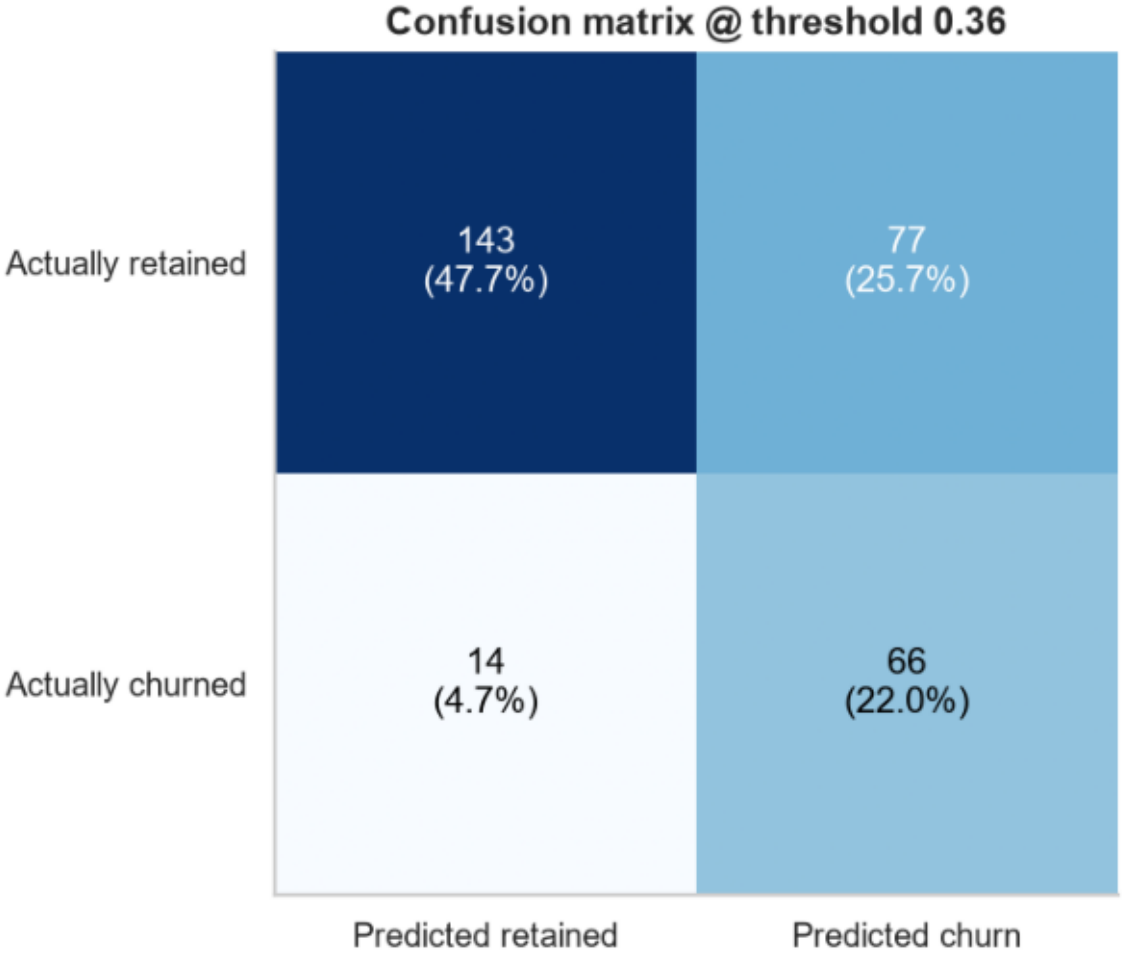

Churn rate by contract type
Overall 26.5%
60%
50%
40%
30%
20%
10%
0%
Churn rate
42.6%
(n=828)
11.6%
(n=311)
2.5%
(n=361)
Month-to-month
One year
Two year
Contract type

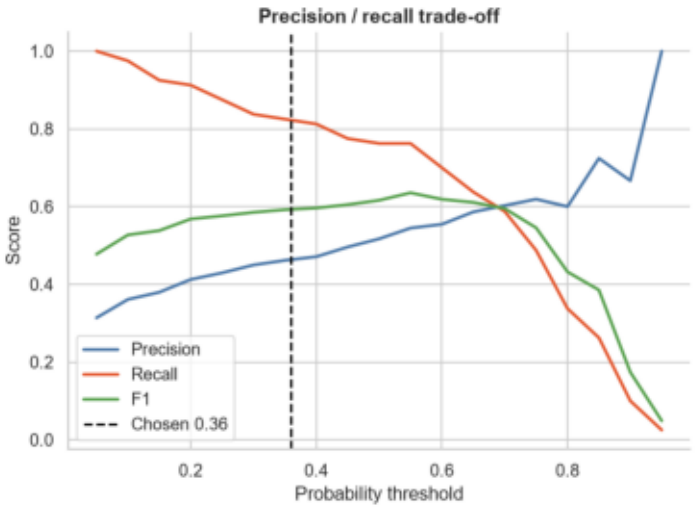

Churn rate by tenure band
Overall 26.5%
80%
70%
60%
50%
40%
30%
20%
10%
0%
Churn rate
56.0%
(n=325)
30.6%
(n=147)
30.8%
(n=227)
20.2%
(n=327)
7.4%
(n=474)
0-6 months
7-12 months
13-24 months
25-48 months
49+ months
Tenure band

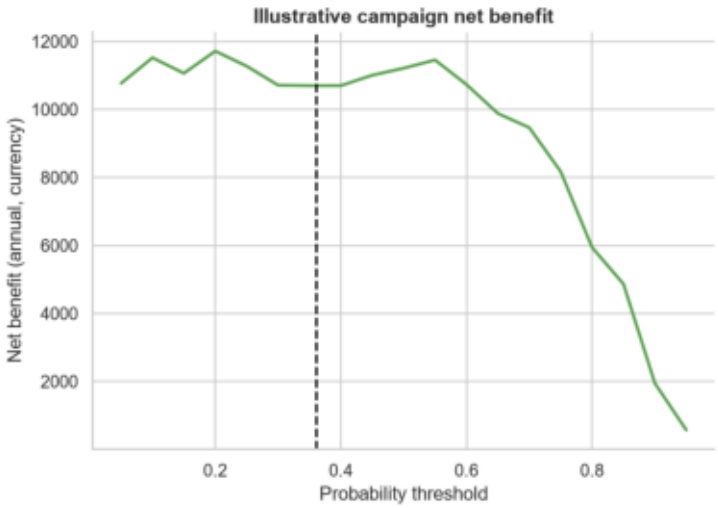

Correlation matrix (numeric variables and churn)
tenure 1.00 0.20 0.81 0.20 0.44 -0.01 0.46 0.29 -0.39
MonthlyCharges 0.20 1.00 0.63 1.00 0.85 -0.02 0.54 0.72 0.18
TotalCharges 0.81 0.63 1.00 0.63 0.77 -0.04 0.66 0.59 -0.22
avg_monthly_spend 0.20 1.00 0.63 1.00 0.84 -0.10 0.54 0.72 0.18
total_services 0.44 0.85 0.77 0.84 1.00 -0.03 0.85 0.76 -0.03
charge_delta -0.01 -0.02 -0.04 -0.10 -0.03 1.00 -0.02 -0.01 -0.00
protection_count 0.46 0.54 0.66 0.54 0.85 -0.02 1.00 0.44 -0.17
streaming_count 0.29 0.72 0.59 0.72 0.76 -0.01 0.44 1.00 0.06
ChurnFlag -0.39 0.18 -0.22 0.18 -0.03 -0.00 -0.17 0.06 1.00
tenure
MonthlyCharges
TotalCharges
avg_monthly_spend
total_services
charge_delta
protection_count
streaming_count
ChurnFlag
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

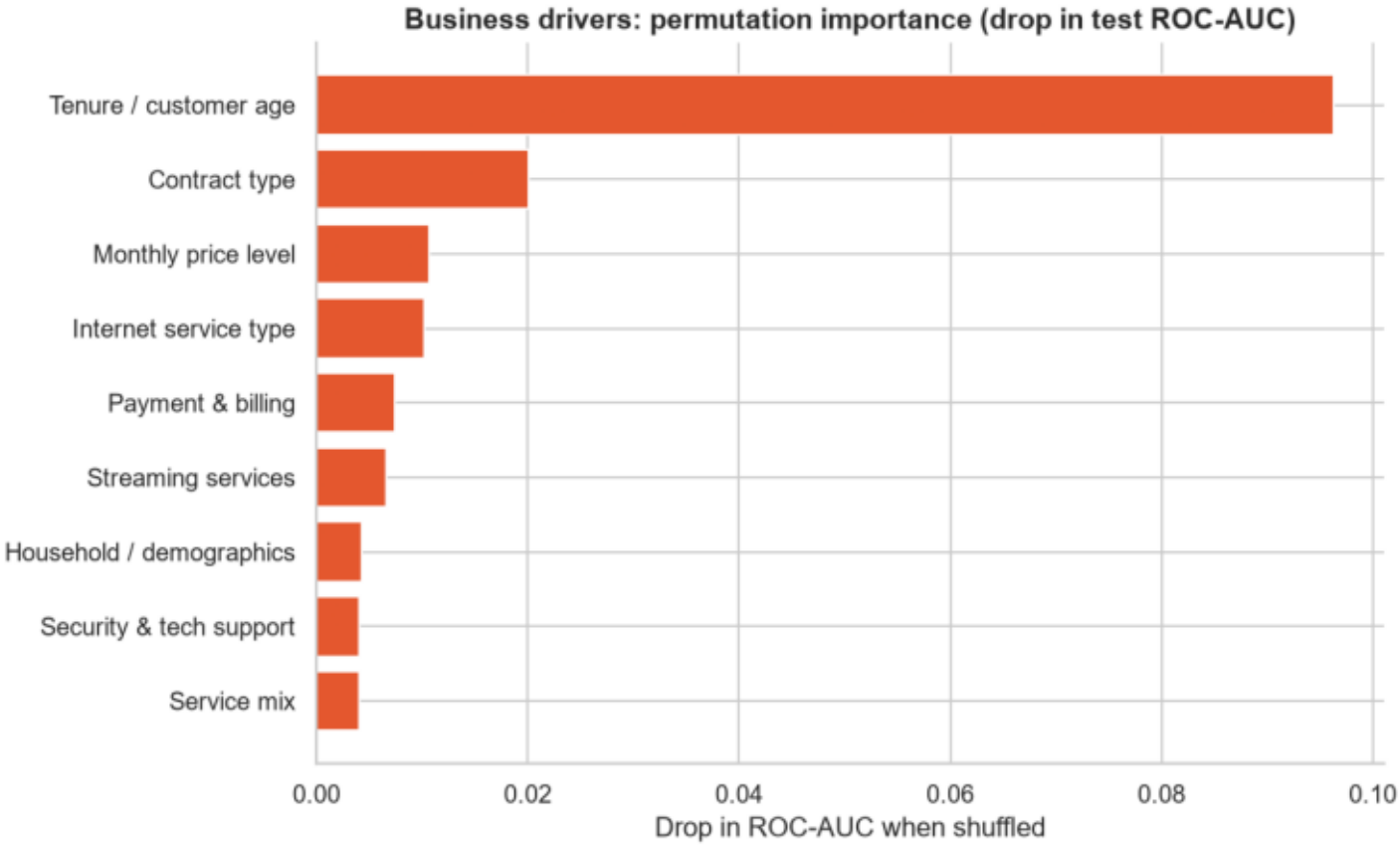

ROC curve (test set)
True positive rate (recall)
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
Model (AUC = 0.825)
Random guess

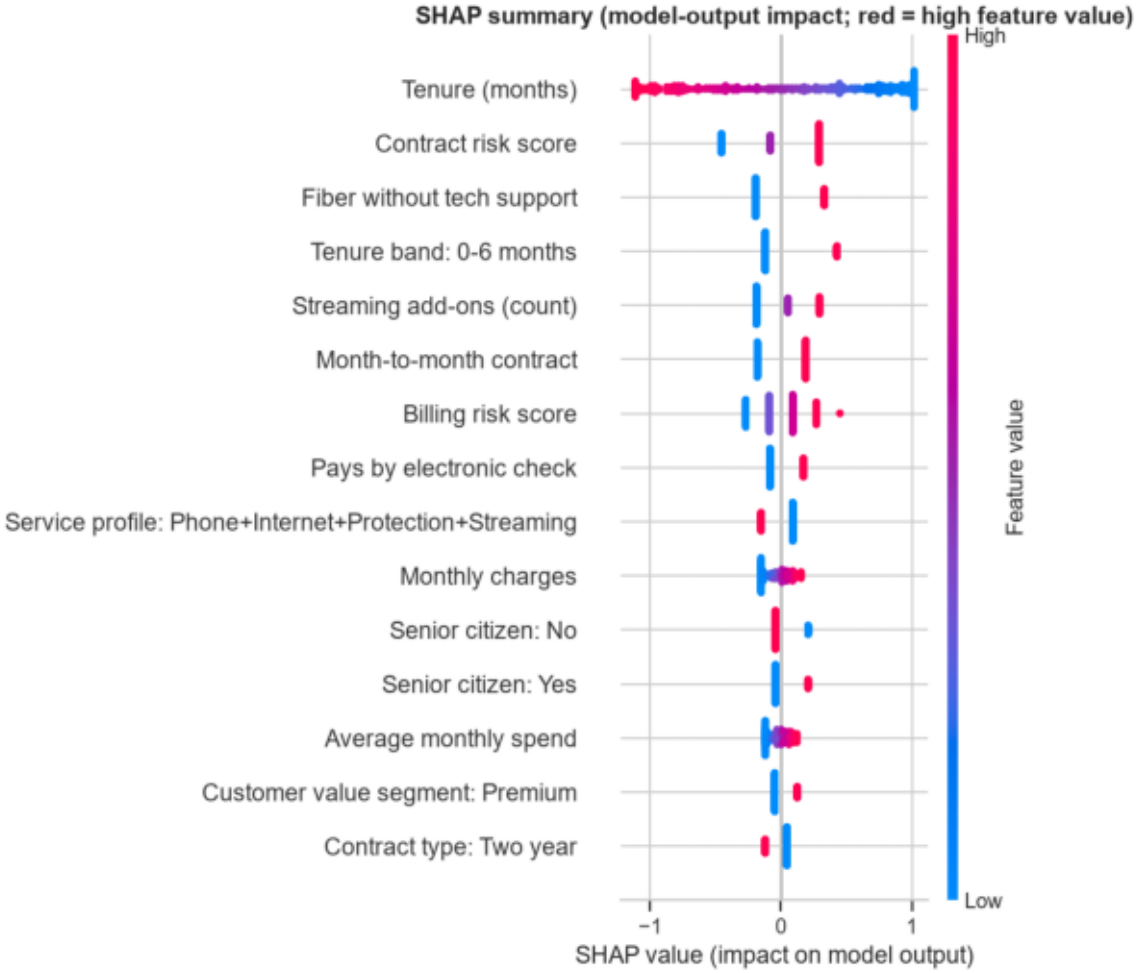

Precision-recall curve (test set)
Model (AP = 0.609)
Baseline (churn rate 26.7%)
Chosen threshold 0.36
Precision
Recall
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.0
0.2
0.4
0.6
0.8
1.0


Confusion matrix @ threshold 0.36
Actually retained
Actually churned
143
(47.7%)
77
(25.7%)
14
(4.7%)
66
(22.0%)
Predicted retained
Predicted churn


Precision / recall trade-off
Score
Probability threshold
Precision
Recall
F1
Chosen 0.36


Illustrative campaign net benefit
Net benefit (annual, currency)
Probability threshold


Business drivers: permutation importance (drop in test ROC-AUC)
Tenure / customer age
Contract type
Monthly price level
Internet service type
Payment & billing
Streaming services
Household / demographics
Security & tech support
Service mix
Drop in ROC-AUC when shuffled


SHAP summary (model-output impact; red = high feature value)
Tenure (months)
Contract risk score
Fiber without tech support
Tenure band: 0-6 months
Streaming add-ons (count)
Month-to-month contract
Billing risk score
Pays by electronic check
Service profile: Phone+Internet+Protection+Streaming
Monthly charges
Senior citizen: No
Senior citizen: Yes
Average monthly spend
Customer value segment: Premium
Contract type: Two year
High
Feature value
Low
−1
0
1
SHAP value (impact on model output)